

CLARITY SemEval 2026 Submission Report

Experiment Forensics Across Granite, Qwen, DeBERTa, RoBERTa, and Ensemble Runs

Workspace: /Users/andrearachetta/Desktop/CLARITY-SemEval-2026

February 14, 2026

Abstract

This report documents the current CLARITY-SemEval-2026 experimentation workspace with a practical focus on (i) model artifacts and metrics, (ii) repository provenance and git state, and (iii) submission-readiness of generated files. The analysis covers Granite, Qwen, DeBERTa, RoBERTa, and ensemble-related assets found in notebooks, JSON files, trainer checkpoints, and folder structures. The repository behaves as an experiment staging area: most model work is currently untracked by git. Consequently, this report emphasizes file-level evidence and explicit paths.

1 Scope and Evidence Collection

Target requested by user. Produce a detailed SemEval-style report with comprehensive tables, focusing on git changes and experiment/model artifacts (especially Granite-related submission workflow), including results recoverable from notebooks, JSON files, and model folders.

Important naming note. No folder or file named `GraniteClaritySubmission` was found. Granite submission logic appears in:

- `GraniteClarityEvaluation.ipynb`
- `GraniteClarityEvaluationcopy.ipynb`
- `GraniteClarityEvaluationcopy2.ipynb`

2 Repository Provenance and Git State

Interpretation. The model experimentation work is primarily filesystem-level (untracked artifacts), not commit-level. This affects reproducibility and diff-based narrative writing.

3 Artifact Footprint by Model Family

Item	Value	Evidence
Current branch	main	git branch --show-current
Commit count in current history	17	git rev-list --count HEAD
Tracked modified files	2	.DS_Store, dataset/.DS_Store
Untracked entries in working tree	27	git status --short
Experiment paths with git history	none	git log -- <experiment path> returns empty for Granite/Qwen/DeBERTa/RoBERTa folders

Table 1: Git provenance snapshot (February 14, 2026).

Family	File Count	Top Extensions	Representative Artifacts
Granite	23	.json, .py, .ipynb, .md	GraniteClarityEvaluation.ipynb; GraniteClarityEvaluationcopy2.ipynb; train_granite_rationale.py; granite_clarity_finetuned/checkpoint-60/trainer_state.json
Qwen	3	.py, .ipynb	QwenTuning.ipynb; train_qwen.py; qwen_classifier.py
DeBERTa	3	.json	deberta/model_evaluation_results_qevasion.json; deberta/model_evaluation_resultscopy.json; deberta/model_evaluation_results.json
RoBERTa	85	.json, .txt, .bin, .yaml, .csv	roberta/roberta-ensemble-model-1/checkpoint-388/trainer_state.json; roberta/roberta-ensemble-model-2/checkpoint-388/trainer_state.json; roberta/train_clarity_ensemble.py; roberta/roberta_ensemble_exported/roberta-ensemble-model-1/model.onnx
Ensemble	88	.json, .yaml, .csv, .py	clarity_ensemble/ensemble_roberta-base-20260201_134652/fold-1/logs/version_0/metrics.csv; ensemble_train_results.json; evaluate_ensemble.py

Table 2: Artifact footprint (substring-based family mapping; families overlap).

4 Data and Split Evidence

Dataset Evidence	Count	Notes
dataset/QAEvasion.csv	3448 rows	Matches train_size=3448 in ensemble config files
dataset/clarity_task_evaluation_dataset.csv	237 rows	Matches submission pickle length (237)
QEvason test split (notebook loader output)	308 rows	Reported by Granite notebook logs before balancing
Balanced Granite evaluation subset	2 69 rows	23 per class in notebook logs (Direct Reply, Direct Non-Reply, Indirect)

Family	Core Configuration	Source
Granite	Base model ibm-granite/granite-3.2-2b-instruct-bnfig.json LoRA: r=8, alpha=32, dropout=0.05; target modules in- clude q_proj/k_proj/v_proj/o_proj and MLP projections.	granite_clarity_finetuned/adapter_
Granite training checkpoint	global_step=60, num_train_epochs=2; training loss decreases from 1.6479 to 1.2488; no stored eval metrics in checkpoint state.	granite_clarity_finetuned/ checkpoint-60/trainer_state.json
Qwen	Training defaults: model unsloth/Qwen3-8B-bnb-4bit, mode classification_head generative, max_epochs=3, batch_size=4, lr=1e-4; LoRA enabled in classifier implementation.	train_qwen.py, qwen_classifier.py
DeBERTa	QEvasion eval file uses microsoft/deberta-v3-base, threshold 0.5; legacy bi- nary evaluation file references lxyuan/distilbert-base-multilingual-cased-sentiments-student.	deberta/model_evaluation_results_ qevasion.json, deberta/model_ evaluation_resultscopy.json
RoBERTa / Clarity en- semble	roberta-base, PEFT enabled, lora_r=16, lora_alpha=32, dropout=0.2, label_smoothing=0.1, lr=2e-5, epochs=5, num_folds=3.	clarity_ensemble/ensemble_ roberta-base_*/config.json, clarity_ensemble/**/hparams.yaml

Table 4: Recovered configuration-level information by model family.

6 Consolidated Quantitative Results

Family	Artifact	Split	n	Acc	F1 (macro)	Prec	Recall	ROC-AUC
Granite	GraniteClarityEvaluation.ipynb (cell 12 output)	Randomly split test	69	0.4638	0.4282	–	–	–
Granite	GraniteClarityEvaluation.ipynb (cells 26–28 outputs)	Balanced test	69	0.3333	0.1667	0.1111	0.3333	–
DeBERTa	deberta/model_evaluation_results_qevasion.json	QEvasion test	308	0.4156	0.3878	0.2651	0.7215	0.54

Family	Artifact	Split	n	Acc	F1 (macro)	Prec	Recall	ROC-AUC
DeBERTa (binary legacy)	deberta/model_ evaluation_ resultscopy.json	Binary eval	200	0.7100	0.3830	0.5000	0.3103	–
RoBERTa fold run	clarity_ensemble/ ensemble_ roberta-base_ 20260201_134652/ fold-1/logs/version_ 0/metrics.csv	Test	308	0.4253	0.3522	0.3683	0.4314	–
RoBERTa fold run	clarity_ensemble/ ensemble_ roberta-base_ 20260201_134652/ fold-1/logs/version_ 0/metrics.csv	Best val	–	0.3913	0.3740	0.3985	0.4717	–

Table 5: Primary comparable metrics across model families from local files and notebook outputs.

Legacy RoBERTa checkpoint diagnostics (not used in comparable ranking)

Checkpoint Artifact	Seed	Epochs	eval_macro_f1	Interpretation
roberta/ roberta-ensemble-model-1/ checkpoint-388/trainer_state. json	42	2	1.0000	Internal checkpoint eval; likely non-comparable to fold metrics.
roberta/ roberta-ensemble-model-2/ checkpoint-388/trainer_state. json	123	2	1.0000	Internal checkpoint eval; likely non-comparable to fold metrics.
roberta/ roberta-ensemble-model-3/ checkpoint-388/trainer_state. json	456	1	0.1786	Different training duration and behavior from model-1/2.

Table 6: Legacy HF Trainer checkpoint diagnostics. Seeds/epochs recovered from training_args.bin artifacts.

7 Granite Notebook Deep-Dive

Additional evidence from `GraniteClarityEvaluation.ipynb` confusion matrix:

$$\begin{bmatrix} 0 & 0 & 23 \\ 0 & 0 & 23 \\ 0 & 0 & 23 \end{bmatrix}$$

Granite Notebook Run	Overall Acc	Macro F1	F1 Direct Reply	F1 Direct Non-Reply
GraniteClarityEvaluationcopy2.ipynb	0.4638	0.4282	0.6538	0.2308
GraniteClarityEvaluation.ipynb	0.3333	0.1667	0.0000	0.0000

Table 7: Granite evaluation metrics extracted from notebook output cells.

which confirms one-label collapse in that run.

8 DeBERTa and Ensemble JSON Results

8.1 DeBERTa result files

File	Examples	Key Signals
deberta/model_evaluation_results_qevasion.json	308	3-class metrics present (Acc/F1/Prec/Recall/ROC-AUC).
deberta/model_evaluation_resultscopy.json	200	Binary metrics with full prediction vectors; older/alternate configuration.
deberta/model_evaluation_results.json	237	Submission-like prediction counts only, no quality metrics.

Table 8: DeBERTa result-file semantics.

8.2 Ensemble train summary file

Entry in ensemble_train_results.json	Acc	Prec (macro)	Recall (macro)	F1 (macro)
model_1_train	0.5916	0.1972	0.3333	0.2478
model_2_train	0.5916	0.1972	0.3333	0.2478
model_3_train	0.3715	0.3074	0.3356	0.2558
ensemble_average_train	0.5916	0.1972	0.3333	0.2478

Table 9: Ensemble summary JSON values (train split only).

9 RoBERTa/Ensemble Run-Health Audit

Run Directory	Config	Metrics CSV	Fold Dirs	Status
20260201_132359	yes	yes	only fold-1	Full training/val log exists; no test line recorded in CSV.
20260201_134652	yes	yes	only fold-1	Most complete run in this folder family; includes one test metric row.
20260201_135710	yes	yes	only fold-1	Truncated (3 rows total).
20260201_185917	yes	yes	only fold-1	Partial (8 rows total).
20260202_135909	yes	no	only fold-1	Config + hparams exist; no metrics CSV written.

Table 10: Run completeness for the clarity_ensemble roberta-base run directories.

Critical reproducibility mismatch. Configs specify num_folds=3, but every run stores only fold-1 artifacts in this workspace snapshot.

Provenance of ensemble_20260201_134652_valbest

The identifier ensemble_20260201_134652_valbest is a post-hoc report label (not a file name). It is defined as:

$$\arg \max_{r \in R} r[\text{val/f1_macro}]$$

where R is the set of rows in: clarity_ensemble/ensemble_roberta-base_20260201_134652/fold-1/logs/version_0/metrics.csv with non-empty val/f1_macro.

Step	Epoch	val/acc	val/f1_macro	val/loss
71	0	0.1035	0.0625	1.1725
143	1	0.1200	0.0827	1.1637
215	2	0.2730	0.2760	1.1507
287	3	0.3548	0.3462	1.1405
359	4	0.3913	0.3740	1.1365

Table 11: Validation checkpoints used to select ensemble_20260201_134652_valbest.

Pipeline-wise, this run is created by:

1. Writing run config and timestamp (roberta/train_clarity_ensemble.py, lines 618–627).
2. Building stratified folds (roberta/train_clarity_ensemble.py, lines 630–650).
3. Training with checkpoint monitor val/f1_macro (roberta/train_clarity_ensemble.py, lines 517–531).
4. Logging metrics via Lightning CSV logger (roberta/train_clarity_ensemble.py, line 536), yielding the row sequence above.

For the same run, the subsequent test row is logged at step 360, epoch 5, with test/f1_macro=0.3522.

Artifact	Observed Status
QwenTuning.ipynb	Recorded runtime failure: <code>torch_shm_manager ... Permission denied</code> (DataLoader worker). Follow-up <code>NameError: model is not defined</code> .
Untitled3.ipynb	Recorded <code>NameError: model is not defined</code> .
train_qwen.py, classifier.py	qwen_ Training/evaluation code exists with LoRA paths and metric logging, but no saved checkpoint (<code>.ckpt</code>) files found in repository.
Qwen metrics artifacts	No standalone Qwen <code>.json/.csv</code> result file found for final numeric reporting.

Table 12: Qwen status: implementation present, but run outputs are incomplete in current workspace.

10 Qwen Experiment Audit

11 Submission Artifact Audit

Artifact	Size/Count	Audit Notes
clarity/clarity_submission.pickle	237 predictions	Tuple format valid (indices + labels). Label distribution: 126 Clear Reply, 111 Non-Clear / Ambiguous Reply.
clarity/clarity_codebench_submission.csv	237 rows	All rows have <code>prediction=2</code> , <code>label=2</code> . This appears degenerate and should not be treated as a final competitive submission without validation.
generate_clarity_submission.py	script	Implements submission generation pipeline via <code>SimpleClarityClassifier</code> ; output label mapping differs from 3-class CLARITY naming used in Granite notebooks.
GraniteClarityEvaluation*.ipynb	notebook logic	Writes <code>clarity_submission_granite.pickle</code> in code cells, but file is not present in current workspace snapshot.

Table 13: Submission-related files and consistency checks.

12 Key Findings

1. **Best recoverable Granite metric: Acc 0.4638, Macro-F1 0.4282** from `GraniteClarityEvaluationcopy.ipynb` on the balanced 69-sample subset.
2. **Best complete RoBERTa fold test metric in clarity_ensemble: Acc 0.4253, Macro-F1 0.3522** from run 20260201_134652.

3. **Best DeBERTa 3-class JSON metric: Acc 0.4156, Macro-F1 0.3878** on 308 examples.
4. **Qwen runs are not yet reportable** due notebook/runtime failures and absence of saved checkpoints/results.
5. **Git provenance is weak for experiments** because nearly all model artifacts are untracked; formal paper claims should cite filesystem evidence and dates explicitly.

13 Recommended Finalization Plan for SemEval Submission Write-Up

1. Promote the complete experiment artifacts (at minimum: configs, metrics CSV/JSON, and selected notebooks) into tracked git commits so paper tables are reproducible from commit hashes.
2. Normalize one canonical label space (**Direct Reply**, **Direct Non-Reply**, **Indirect**) across Granite/DeBERTa/RoBERTa/legacy binary artifacts before final leaderboard submission.
3. Re-run Qwen with fixed DataLoader settings (`num_workers=0` and verified shared-memory permissions), then export a dedicated metrics JSON comparable with other families.
4. Complete true 3-fold outputs for `clarity_ensemble` or explicitly document single-fold limitation.
5. Generate one final, validated submission file from the selected best model and store it with a dated manifest (model, commit hash, metrics source paths).